

NASSAU CANDY DISTRIBUTOR

Factory-to-Customer Shipping Route Efficiency Analysis

Business Analyst Internship Project Report

BUSINESS ANALYST POSITIONING

This report converts the project brief and uploaded CSV into a stakeholder-ready analytical framework: business problem, validated data model, KPI definitions, route diagnostics, risks, recommendations, and Streamlit dashboard requirements.

Shipment lines	Unique orders	Sales	Gross profit
10,194	8,549	\$141,783.63	\$93,442.80

Customers	Products	States / provinces	Gross margin
5,044	15	59	65.9%

Prepared from the user-provided project brief, factory coordinates, product-factory mapping, and Nassau Candy Distributor CSV.

1. Executive Summary

The project objective is to make factory-to-customer shipping performance visible at route, state, region, and shipping-mode level so logistics decisions can be based on evidence rather than intuition. The analysis uses two grains: shipment-line data for financial totals and unique-order data for route and lead-time metrics. This avoids overstating route volume when a single order contains multiple product lines.

KEY MANAGEMENT MESSAGE

The dataset contains 10,194 shipment lines and 8,549 unique orders. The strongest business signals are volume concentration and data quality. Lot's O' Nuts accounts for 55.7% of US orders, Standard Class represents 59.9% of US orders, and California is the largest customer state by orders and sales. However, the recorded shipment dates create lead times of 904 to 1642 days, so literal delivery-speed conclusions are not operationally reliable until the Ship Date field is validated.

What is decision-ready now

- Volume concentration: the Pacific region contains 2,698 US orders and \$46,301.53 in sales.
- Factory concentration: Lot's O' Nuts contributes \$76,340.15 in sales and \$52,771.05 in gross profit.
- Shipping mix: Standard Class is the dominant mode with 5,026 US orders and \$85,490.35 in sales.
- Geographic concentration: California has 1,686 US orders and \$27,917.40 in sales.

What is not decision-ready yet

- Average lead time, delay frequency, and route efficiency scores cannot be interpreted as real-world delivery durations because all recorded lead times exceed 900 days.
- The dataset includes 200 Canadian shipment lines although the project scope is US regions; Canadian rows are excluded from US route rankings in this report and should be retained as a separate geography in the dashboard.
- The dashboard should display a data-quality banner whenever the shipment-date anomaly is present rather than silently presenting misleading delivery KPIs.

2. Business Problem and BA Objectives

The underlying business problem is not simply "find the fastest route." A logistics analyst needs to answer five questions: where demand is concentrated, where performance varies, whether variability is stable, how ship mode changes the service profile, and whether the data is trustworthy enough to support action.

Stakeholder	Primary question	Decision supported
Stakeholder	Primary question	Decision supported
Logistics / Operations	Which factory-state routes need investigation?	Network and carrier review
Customer Service	Where could delays affect customer experience?	Proactive exception management
Finance	Which high-volume routes carry significant sales/profit?	Prioritization by business exposure
Data / IT	Can lead-time KPIs be trusted?	Source-system / ETL remediation
Management	What should be monitored weekly?	KPI governance and accountability

Analytical objectives

- Establish a repeatable, auditable route definition: Factory -> Customer State and Factory -> Customer Region.
- Use unique Order ID as the volume grain for route metrics; use shipment-line records for financial aggregation.
- Benchmark average lead time and variability only after date validity is confirmed.
- Prioritize high-volume routes for investigation using both operational and financial exposure.
- Translate analysis into a Streamlit dashboard with filters, drill-down, and data-quality controls.

Success criteria

BA SUCCESS CRITERION

A manager should be able to select a date range, state/region, and ship mode; identify the highest-volume routes; inspect lead-time variation; and trace any route KPI back to order-level records without needing a separate analyst.

3. Dataset Profiling and Data Quality

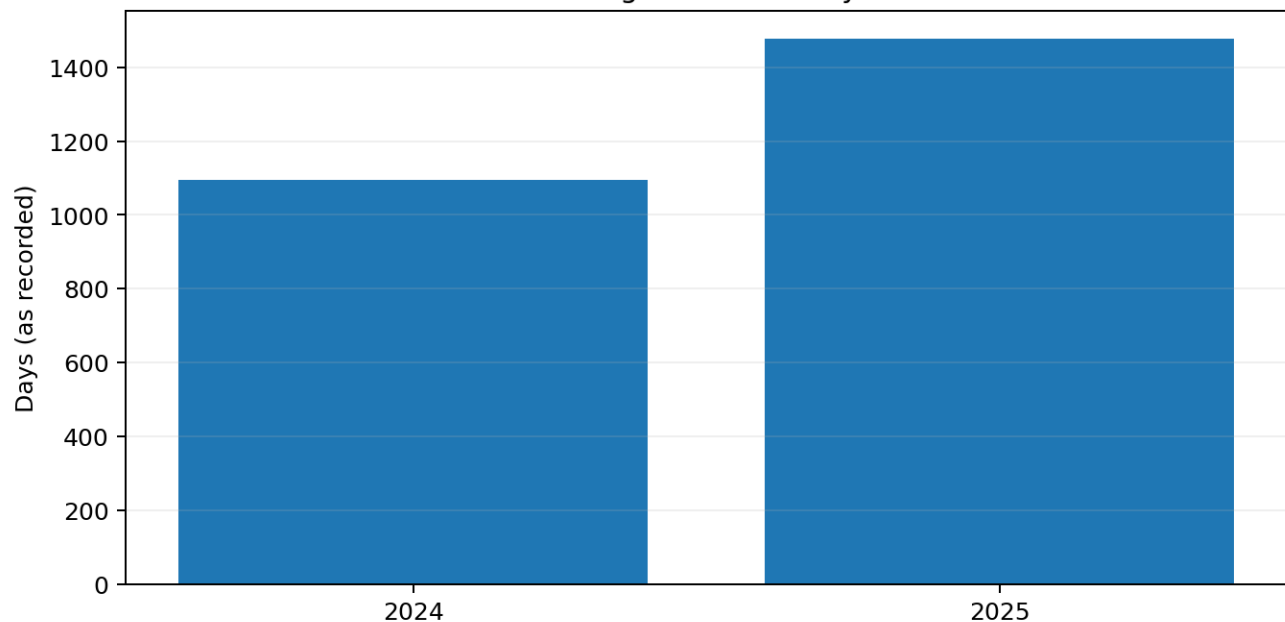
The CSV contains 18 fields covering orders, shipment dates, customer geography, products, sales, units, cost, and gross profit. The profile shows complete populated fields, a one-to-many relationship between orders and shipment lines, and a major temporal consistency issue that must be handled explicitly.

Profile item	Result	BA interpretation
Profile item	Result	BA interpretation
Rows / shipment lines	10,194	Line-level facts; useful for financial totals
Unique orders	8,549	Correct grain for route volume and lead-time analysis
Unique customers	5,044	Useful for customer concentration analysis
Products	15	All mapped to a factory using the project specification
States / provinces	59	Includes US plus Canada
Missing values	0 across all 18 fields	Strong completeness at file level
Duplicate Row ID	0	No duplicate row identifiers detected
Duplicate Order ID	1,645 extra line items	Expected multi-line order behavior; not a data defect by itself
Canada rows	200 (1.96%)	Outside stated US scope; isolate in reporting
Product -> Factory mapping	0 unmapped products	Mapping is complete for provided 15-product list

CRITICAL DATA-QUALITY ISSUE

Order Date spans 02-Jan-2024 to 31-Dec-2025, while Ship Date spans 30-Jun-2026 to 28-Jun-2030. As recorded, every shipment has a lead time between 904 and 1642 days. All 10,194 rows are therefore above one year. This is a semantic/data-source problem, not a normal logistics pattern.

Recorded Average Lead Time by Order Year



The increase in recorded lead time from 2024 orders to 2025 orders is a strong signal that the date fields are not comparable as actual shipment elapsed time.

4. Data Model, Cleaning Rules, and Analytical Grain

The analysis uses a simple star-like model. This is important because a business analyst should prevent multi-line orders from inflating route counts while still preserving all financial line items.

Table / grain	Key	Purpose
Table / grain	Key	Purpose
Shipment Line Fact	Row ID	Sales, units, cost, gross profit, product-level detail
Order Fact	Order ID	One record per order; lead time, route, customer and ship mode
Factory Dimension	Factory	Factory coordinates and product-origin mapping
Product Dimension	Product ID / Product Name	Product, division, factory
Geography Dimension	State / Region	US/Canada geography and regional roll-up
Ship Mode Dimension	Ship Mode	Shipping service classification

Cleaning and validation rules

- Parse dates explicitly in DD-MM-YYYY format; reject unparseable records.
- Create Lead Time = Ship Date - Order Date at unique Order ID grain.
- Flag any negative or zero lead time for investigation.
- Flag lead times that exceed an agreed operational upper bound after the business confirms expected service levels.
- Standardize geography; preserve original values for auditability.
- Map products to factories using the provided product-factory correlation table; fail the pipeline if a product is unmapped.
- Use unique orders for route volume and lead-time calculations; use shipment lines for Sales, Cost, Gross Profit, and Units.

Recommended data-quality fields in the dashboard

Field	Rule	Dashboard behavior
Field	Rule	Dashboard behavior
Date Validity Flag	Order and Ship dates pass source validation	Show green status
US Scope Flag	Country/Region = United States	Used for US route KPIs
Route Sample Flag	Unique orders >= 20	Eligible for route benchmarking
Lead Time Validity Flag	Lead time within agreed service bounds	Exclude or warn when invalid

5. KPI Framework

KPI	Definition	Grain	Use
KPI	Definition	Grain	Use
Shipping Lead Time	Ship Date - Order Date	Order	Core service metric; currently blocked by date issue
Average Lead Time	Mean lead time	Route / state / region / mode	Central tendency
Median Lead Time	Median lead time	Route / state / region / mode	Robust view of skew
Lead Time Variability	Std. deviation or IQR	Route	Reliability / consistency
Route Volume	Unique Order ID count	Route	Exposure / demand concentration
Delay Frequency	% of orders above approved threshold	Route / mode / region	Exception rate; threshold is configurable
Route Efficiency Score	Normalized lead-time score	Eligible route	Relative benchmarking after date validation
Gross Profit Margin	Gross Profit / Sales	Factory / state / route	Financial exposure and economics

Diagnostic route score used in this report

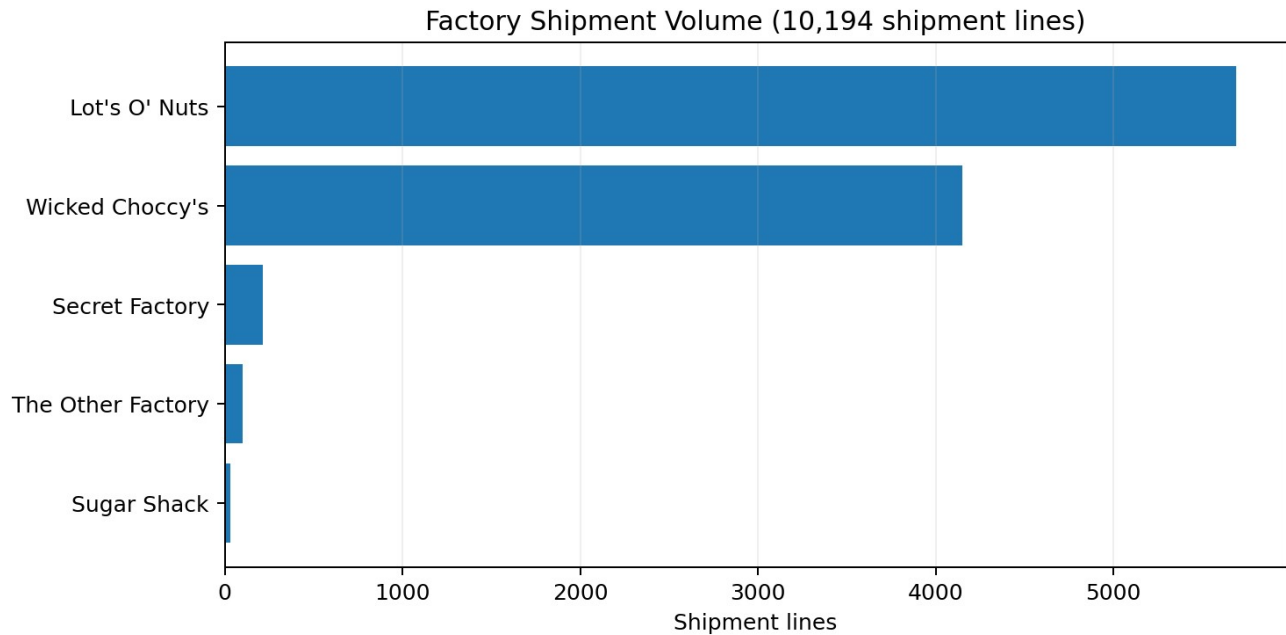
Because the calendar dates are not operationally trustworthy, the route score is presented only as a diagnostic ranking. For US routes with at least 20 unique orders:

Diagnostic Efficiency Score

Score = $100 \times (\text{maximum route average lead time} - \text{route average lead time}) / (\text{maximum route average lead time} - \text{minimum route average lead time})$. A higher score means a lower recorded average lead time within the eligible comparison set. This score must not be treated as an operational SLA score until the date issue is corrected.

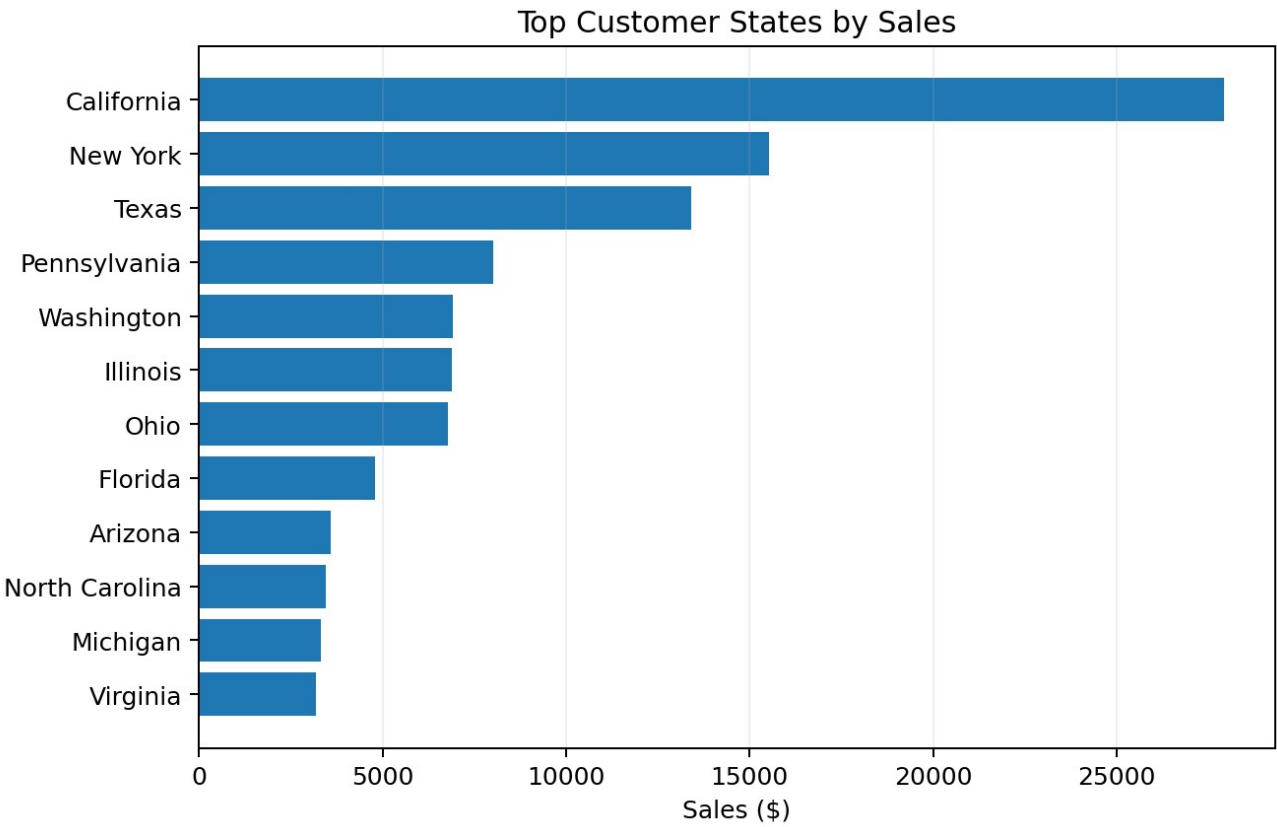
6. Exploratory Data Analysis - Business Volume

The file represents \$141,783.63 of sales, \$93,442.80 gross profit, and 38,654 units. The overall gross margin is 65.9%. Business exposure is highly concentrated in two factories and a small number of customer states.



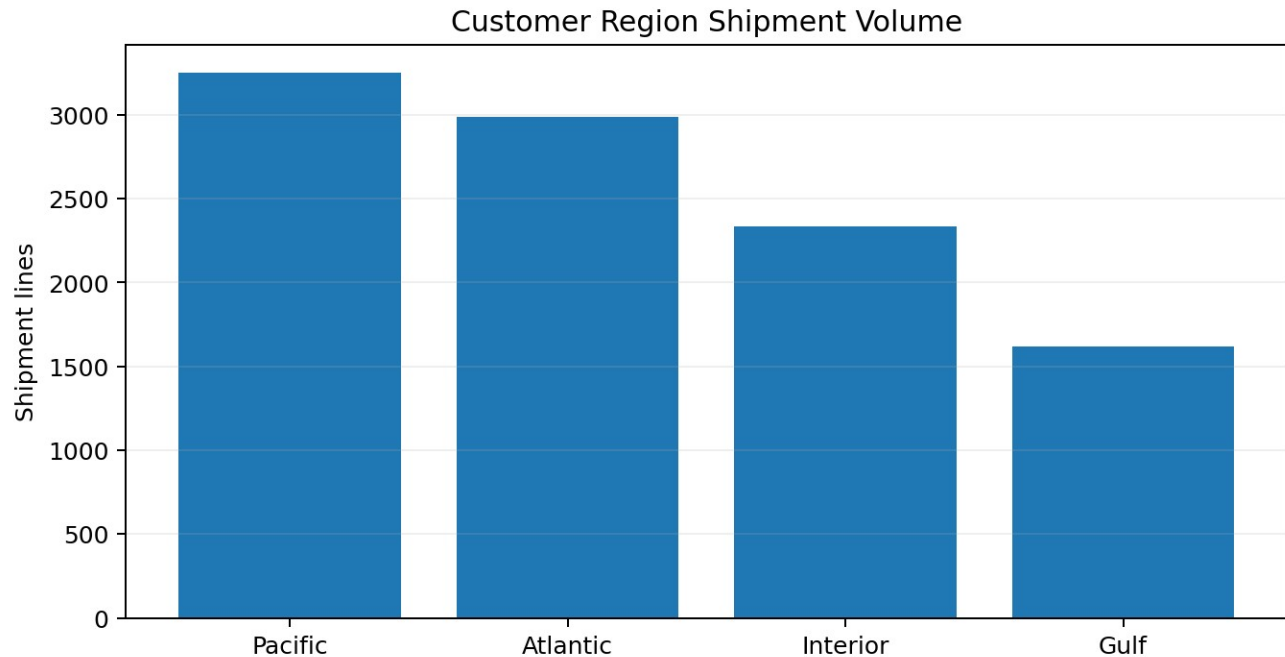
Lot's O' Nuts is the largest factory by shipment-line volume, followed by Wicked Choccy's.

Factory	US orders	Sales	Gross profit	Sales share
Lot's O' Nuts	4,669	\$76,340.15	\$52,771.05	53.8%
Wicked Choccy's	3,380	\$55,352.75	\$36,053.57	39.0%
Secret Factory	209	\$8,587.50	\$4,344.70	6.1%
The Other Factory	98	\$1,282.25	\$152.25	0.9%
Sugar Shack	33	\$220.98	\$121.23	0.2%



California, New York, and Texas are the three largest states by sales in the uploaded data.

7. Geographic Analysis and Bottleneck Framework



Customer-region distribution by shipment-line volume. Pacific carries the largest share of the observed network volume.

Region	US orders	Order share	Sales	Sales share	Avg lead (recorded days)
Pacific	2,698	32.2%	\$46,301.53	32.7%	1,323.0
Atlantic	2,370	28.3%	\$41,197.24	29.1%	1,324.5
Interior	1,971	23.5%	\$32,037.60	22.6%	1,323.8
Gulf	1,350	16.1%	\$22,247.26	15.7%	1,312.0

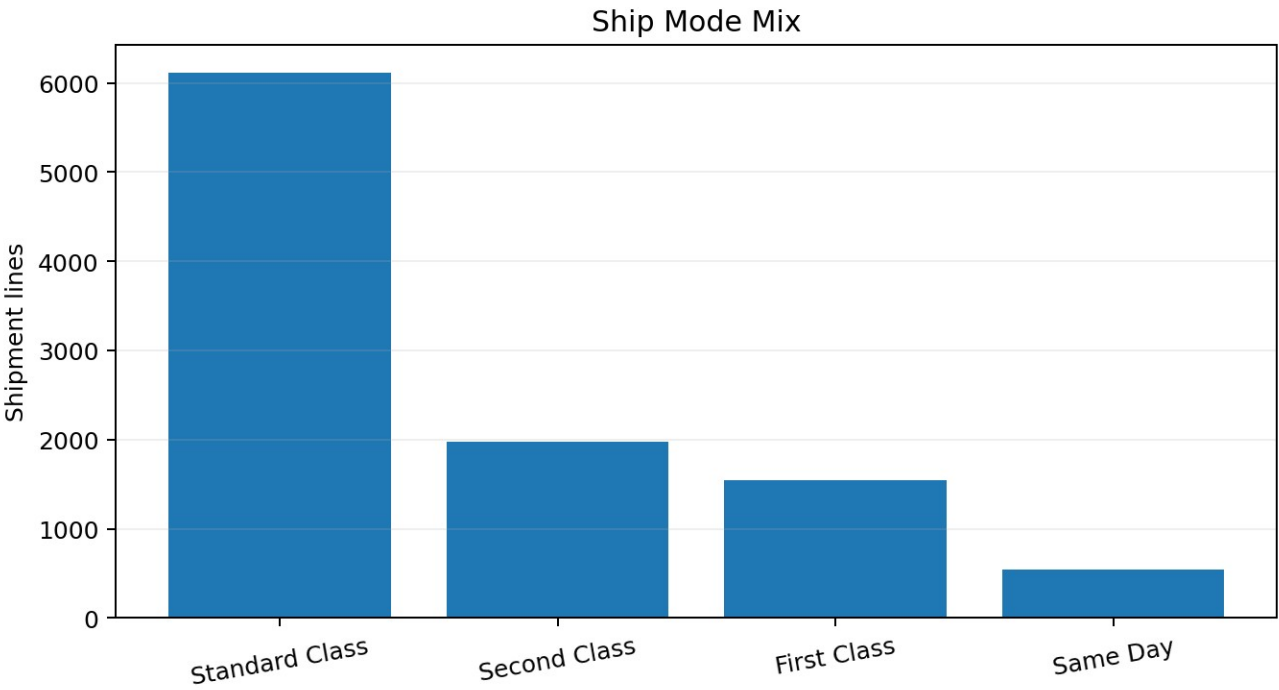
High-exposure states

State	US orders	Sales	Gross profit	Avg lead (recorded)
California	1,686	\$27,917.40	\$18,479.42	1,319.1
New York	942	\$15,541.03	\$10,222.44	1,323.3
Texas	833	\$13,416.09	\$8,909.53	1,312.2
Pennsylvania	487	\$8,027.03	\$5,225.47	1,320.4
Washington	432	\$6,921.15	\$4,566.64	1,364.6
Illinois	433	\$6,898.96	\$4,557.68	1,321.5
Ohio	392	\$6,768.95	\$4,413.03	1,327.9
Florida	319	\$4,804.02	\$3,207.11	1,313.3
Arizona	190	\$3,587.55	\$2,290.11	1,308.0
North Carolina	216	\$3,450.86	\$2,331.16	1,329.1

Bottleneck interpretation

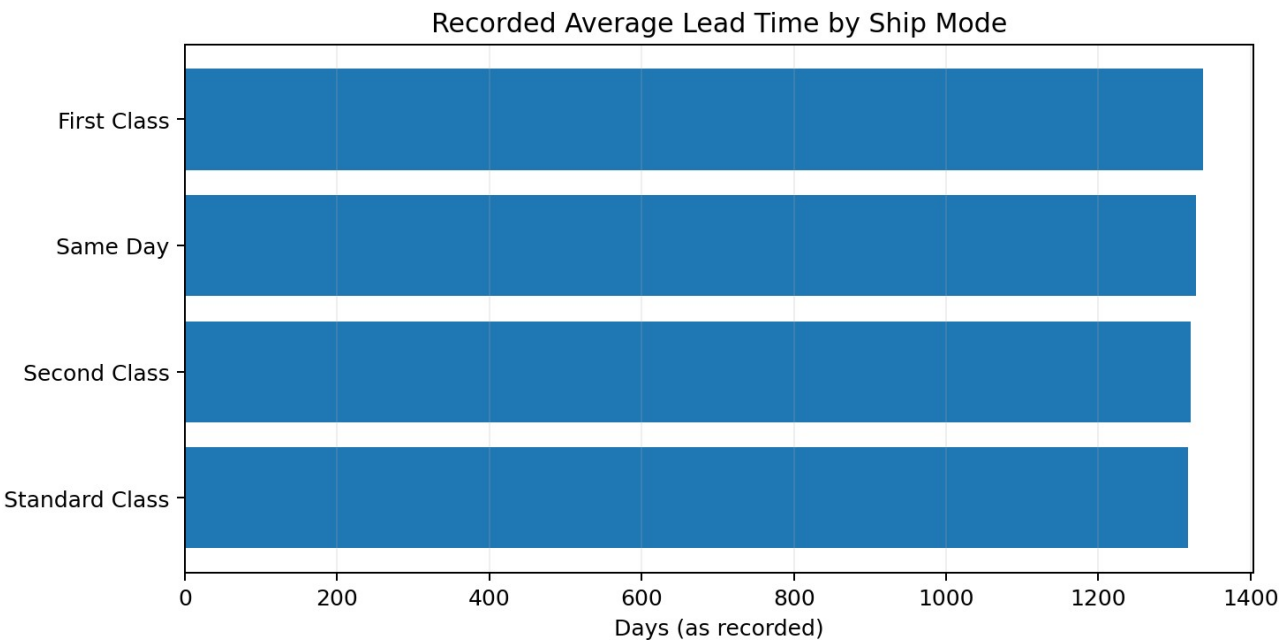
California, New York, Texas, Pennsylvania, Washington, Illinois, Ohio, and Florida represent major business exposure because of volume and sales. They should be the first validation targets after the shipment-date issue is resolved. Their current recorded lead times are not evidence of actual delay performance.

8. Ship Mode Performance Analysis



Standard Class is the dominant service mode in the uploaded file.

Ship mode	US orders	Order share	Sales	Gross margin	Avg lead (recorded)
Standard Class	5,026	59.9%	\$85,490.35	66.0%	1,317.6
Second Class	1,621	19.3%	\$27,860.22	65.7%	1,321.0
First Class	1,302	15.5%	\$21,319.39	65.7%	1,337.2
Same Day	440	5.2%	\$7,113.67	66.1%	1,328.3



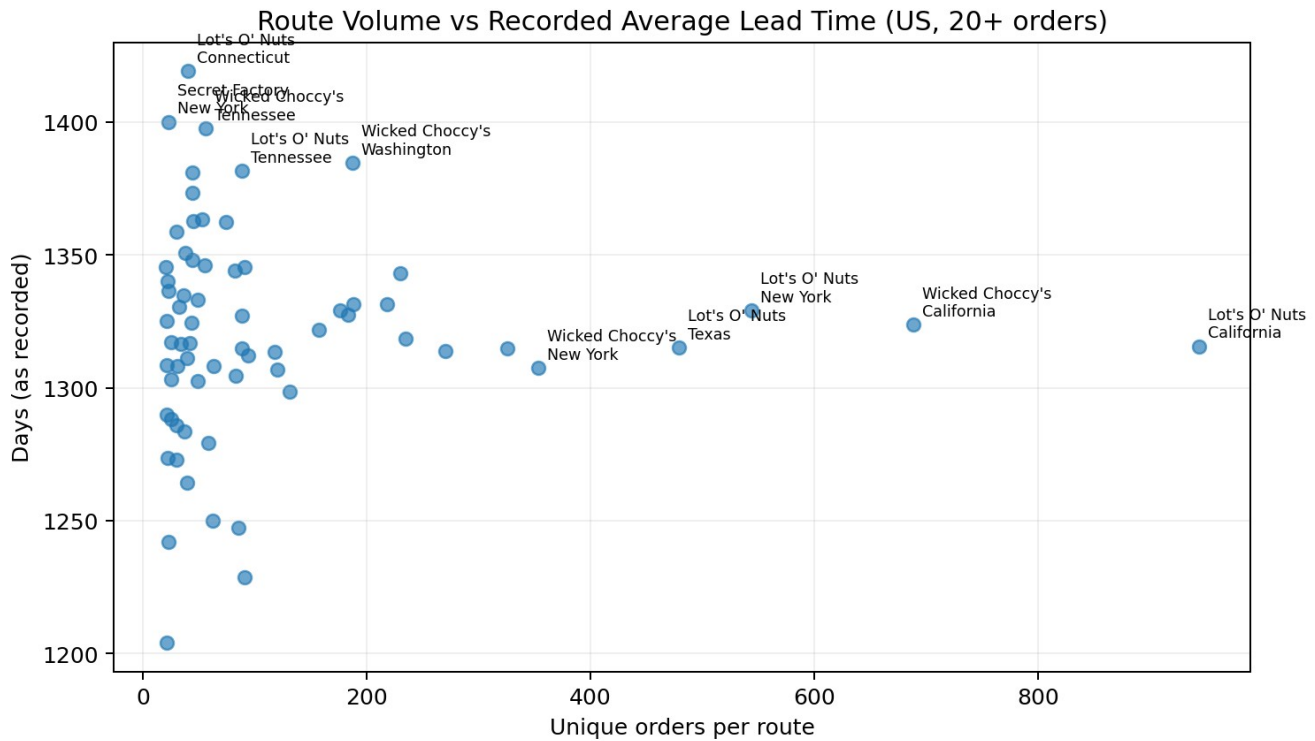
The recorded average lead-time differences by ship mode are small relative to the data anomaly; they should not be interpreted as evidence about service-level superiority.

BA interpretation of the cost-time tradeoff

- Standard Class carries the majority of orders and sales, so even a modest change in service quality has broad operational exposure.
- First Class has a higher recorded average lead time than Standard Class, which contradicts normal expectations for an expedited service and reinforces the date-field concern.
- Do not recommend changing customer shipping policies based on this file alone; first validate actual dispatch and delivery dates and any carrier SLA fields.

9. Route Efficiency Analysis - Diagnostic View

For the route benchmark, I used unique US orders and required a minimum of 20 orders per Factory -> State route. This reduces noise from one-off routes. That leaves 67 eligible routes. The rankings below are deliberately labeled diagnostic because the underlying recorded lead times are not operationally valid.



Route volume and recorded lead time. High-volume routes deserve investigation first because they combine operational exposure with measurable business impact.

Top 10 routes by recorded average lead time (diagnostic)

Route	Orders	Avg lead	Std dev	Diagnostic score
Lot's O' Nuts -> Utah	21	1,204.0	296.9	100.0
Lot's O' Nuts -> Virginia	91	1,228.8	248.8	88.5
Lot's O' Nuts -> Nevada	23	1,241.9	244.5	82.4
Wicked Choccy's -> Virginia	85	1,247.4	257.2	79.8
Lot's O' Nuts -> Oregon	62	1,249.9	254.8	78.7
Lot's O' Nuts -> Delaware	39	1,264.2	257.9	72.0
Wicked Choccy's -> Connecticut	30	1,272.8	272.0	68.0
Wicked Choccy's -> Utah	22	1,273.5	297.4	67.7
Lot's O' Nuts -> Massachusetts	58	1,279.3	260.5	65.0
Wicked Choccy's -> Delaware	37	1,283.7	219.4	63.0

Bottom 10 routes by recorded average lead time (diagnostic)

Route	Orders	Avg lead	Std dev	Diagnostic score
Lot's O' Nuts -> Connecticut	40	1,419.1	259.5	0.0
Secret Factory -> New York	23	1,400.1	261.1	8.8

Wicked Choccy's -> Tennessee	56	1,397.7	234.1	10.0
Wicked Choccy's -> Washington	187	1,384.6	264.8	16.0
Lot's O' Nuts -> Tennessee	88	1,381.6	254.8	17.5
Lot's O' Nuts -> Maryland	44	1,381.0	231.0	17.7
Wicked Choccy's -> Indiana	44	1,373.4	254.1	21.3
Lot's O' Nuts -> New Jersey	53	1,363.5	267.5	25.9
Wicked Choccy's -> Kentucky	45	1,362.6	271.6	26.3
Lot's O' Nuts -> Indiana	74	1,362.3	240.7	26.4

Interpretation boundary

These rankings are useful for testing the dashboard logic, not for operational decisions. The route order is driven by a recorded date span with values above 900 days. Once the business confirms actual shipment dates, rerun this section and replace all diagnostic language with validated delivery KPIs.

10. Variability and Route Prioritization

A business analyst should not select routes only by average. A route with a stable lead time is operationally different from one with the same average but high dispersion. The table below surfaces routes with higher observed standard deviation among routes with at least 20 US orders.

Route	Orders	Avg lead (recorded)	Std dev
Wicked Choccy's -> Minnesota	36	1,334.8	308.6
Wicked Choccy's -> Arkansas	21	1,308.4	302.9
Wicked Choccy's -> Utah	22	1,273.5	297.4
Lot's O' Nuts -> Utah	21	1,204.0	296.9
Lot's O' Nuts -> Arkansas	25	1,303.1	296.7
Lot's O' Nuts -> Louisiana	21	1,289.9	293.8
Lot's O' Nuts -> Mississippi	25	1,288.1	288.5
Secret Factory -> California	31	1,308.3	288.3
Wicked Choccy's -> Missouri	25	1,317.0	284.9
Lot's O' Nuts -> Minnesota	38	1,350.9	283.8

High-volume routes to validate first

Route	Orders	Avg lead (recorded)	Region
Lot's O' Nuts -> California	944	1,315.7	Pacific
Wicked Choccy's -> California	689	1,323.7	Pacific
Lot's O' Nuts -> New York	544	1,329.0	Atlantic
Lot's O' Nuts -> Texas	479	1,315.2	Interior
Wicked Choccy's -> New York	353	1,307.5	Atlantic
Wicked Choccy's -> Texas	326	1,314.7	Interior
Lot's O' Nuts -> Pennsylvania	270	1,313.7	Atlantic
Lot's O' Nuts -> Illinois	235	1,318.6	Interior
Lot's O' Nuts -> Washington	230	1,343.2	Pacific
Lot's O' Nuts -> Ohio	218	1,331.5	Atlantic
Wicked Choccy's -> Pennsylvania	188	1,331.6	Atlantic
Wicked Choccy's -> Washington	187	1,384.6	Pacific

Priority logic

The first operational review list should combine route volume, financial exposure, and validated service performance. Until dates are fixed, the current list should be treated as a data-validation queue rather than a list of proven bottlenecks.

11. Recommendations and Business Actions

Priority	Action	Owner	Expected value
Priority	Action	Owner	Expected value
1	Confirm Ship Date semantics with source/ETL owner and replace planned/forecast dates with actual shipment dates where available.	Data / IT + Logistics	Makes lead-time KPIs trustworthy
2	Implement automated validation: date ranges, negative leads, maximum lead threshold, US/Canada scope, and unmapped products.	Data / IT	Prevents silent KPI corruption
3	Use unique Order ID as route volume and lead-time grain; preserve line-level finance.	Analytics	Prevents route-volume overstatement
4	After date correction, benchmark routes using average, median, P90 lead time, variability, and volume.	Business Analytics	Balances speed, reliability, and exposure
5	Start review with high-volume states such as California, New York, Texas, Pennsylvania, Washington, Illinois, and Ohio.	Logistics	Targets the largest business exposure
6	Add a dashboard data-quality banner and drill-through from route -> orders.	BI / Streamlit developer	Builds user trust and traceability

Decision framework after data remediation

- Route efficiency: compare average and P90 lead time against the agreed SLA.
- Reliability: compare variability and delay frequency, not just average.
- Exposure: weight attention by unique order volume, sales, and gross profit.
- Mode tradeoff: compare validated service outcomes against shipping cost once a freight-cost field is available.
- Continuous monitoring: review exceptions weekly and route performance monthly.

12. Streamlit Dashboard Specification

The dashboard should translate the analysis into a practical decision interface. The user should always be able to see the selected filters, KPI grain, data-quality status, and number of orders included.

Page / module	Visuals	Controls	BA acceptance criterion
Page / module	Visuals	Controls	BA acceptance criterion
Executive Overview	KPI cards, order volume, sales, profit, data-quality banner	Date, region, state, ship mode	All KPIs refresh consistently and show sample size
Route Efficiency	Route leaderboard, volume vs lead scatter, score table	State, factory, minimum order threshold	Routes are ranked only when threshold is met
Geographic Shipping Map	State heatmap, region comparison	State/region, ship mode, lead threshold	Map uses validated US state metrics
Ship Mode Comparison	Average/median/P90 lead, volume, sales by mode	Ship mode	Mode comparison is descriptive, not causal
Route Drill-Down	Route summary + order timeline table	Factory, state, order ID	User can trace route KPI to underlying orders
Data Quality	Missing values, date ranges, invalid lead counts	None	Quality status is visible before performance charts

Required filters

- Date range
- Region
- State / Province
- Factory
- Ship Mode
- Lead-time threshold
- Minimum route order count

Recommended dashboard KPI cards

Card	Definition	Data status
Card	Definition	Data status
Unique Orders	Distinct Order ID	Ready
Sales	Sum of Sales	Ready
Gross Profit	Sum of Gross Profit	Ready
Average Lead Time	Mean order lead time	Blocked until dates validated
Delay Frequency	% above threshold	Blocked until dates validated
Route Efficiency Score	Normalized route metric	Blocked until dates validated

13. Research Paper / Methodology Section

Research objective

To evaluate factory-to-customer shipping route efficiency using order, shipment, product, and geographic data and to identify high-exposure routes that require operational attention.

Research questions

Which factory-state routes show different recorded lead-time profiles? How does performance vary by region and ship mode? Which routes combine high demand with weak performance? What data-quality conditions can prevent reliable logistics decisions?

Method

Data validation -> order-grain creation -> factory mapping -> route aggregation -> descriptive EDA -> efficiency benchmarking -> bottleneck prioritization -> dashboard specification.

Unit of analysis

Unique Order ID for route and lead-time metrics; shipment-line rows for financial aggregation.

Limitations

The uploaded file has ship dates years after order dates, so lead-time statistics cannot be treated as real shipping durations. The file also contains 200 Canada rows outside the stated US scope.

Ethical / business-use note

Recommendations are framed as operational hypotheses requiring validation, not as claims of causal impact. No route should be reconfigured solely from this dataset until date semantics are verified.

Research findings summary

- The dataset contains 10,194 line items, 8,549 orders, 5,044 customers, and 15 products.
- Overall sales are \$141,783.63 with gross profit of \$93,442.80 and gross margin of 65.9%.
- Lot's O' Nuts and Wicked Choccy's account for 92.9% of sales combined.
- Standard Class represents 59.9% of US orders.
- California is the largest US customer state by orders (1,686) and sales (\$27,917.40).
- All recorded lead times lie between 904 and 1642 days, so the shipment-date field requires source validation before service-level conclusions are published.

14. Executive / Stakeholder Summary

ONE-PAGE MESSAGE

The analysis can already support volume, geography, factory, product-origin, and financial visibility. It cannot yet support a credible statement about actual delivery speed because the date fields produce 904-1,642 day lead times. The highest-value action is therefore to fix the date semantics and rerun the same repeatable route model, not to manually alter the data.

Area	Current evidence	Action for stakeholder
Area	Current evidence	Action for stakeholder
Network exposure	Pacific is the largest region; California is the largest state by orders and sales.	Use these areas as the first operational validation targets.
Factory exposure	Lot's O' Nuts is the largest factory by order volume and sales.	Prioritize factory-state route review there first.
Shipping mix	Standard Class dominates order volume.	Use it as the baseline service mode after dates are corrected.
Data quality	All rows have complete fields, but lead time is 904-1,642 days.	Escalate date-field validation before publishing SLA KPIs.
Dashboard readiness	Route model and filters are defined.	Implement a data-quality gate before route ranking visuals.

Bottom line

The project is well-suited to a Business Analyst portfolio because it demonstrates more than visualization: it shows problem framing, grain selection, data-quality control, KPI design, route analytics, stakeholder requirements, and a clear path from data issue to business action.

Appendix A - Factory Coordinates

Factory	Latitude	Longitude
Lot's O' Nuts	32.881893	-111.768036
Wicked Choccy's	32.076176	-81.088371
Sugar Shack	48.119140	-96.181150
Secret Factory	41.446333	-90.565487
The Other Factory	35.117500	-89.971107

These coordinates are carried directly from the project specification and can be used to draw factory-origin markers in the Streamlit map.

Appendix B - Product to Factory Mapping

Division	Product Name	Factory
Chocolate	Wonka Bar - Milk Chocolate	Wicked Choccy's
Chocolate	Wonka Bar - Triple Dazzle Caramel	Wicked Choccy's
Chocolate	Wonka Bar - Nutty Crunch Surprise	Lot's O' Nuts
Chocolate	Wonka Bar - Scrumdiddlyumptious	Lot's O' Nuts
Chocolate	Wonka Bar - Fudge Mallows	Lot's O' Nuts
Other	Wonka Gum	Secret Factory
Other	Kazookles	The Other Factory
Other	Lickable Wallpaper	Secret Factory
Sugar	Fizzy Lifting Drinks	Sugar Shack
Sugar	Laffy Taffy	Sugar Shack
Sugar	SweeTARTS	Sugar Shack
Sugar	Nerds	Sugar Shack
Sugar	Hair Toffee	The Other Factory
Sugar	Everlasting Gobstopper	Secret Factory
Sugar	Fun Dip	Sugar Shack

Appendix C - Reproducibility Notes

- Date parsing is explicitly day-first using the source format DD-MM-YYYY.
- US route metrics exclude Country/Region = Canada. Canadian records are retained in the raw dataset for separate analysis.
- Route volume is based on unique Order ID because orders can contain multiple product lines.
- Financial totals use all shipment-line rows because Sales, Cost, Gross Profit, and Units are line-level measures.
- Route ranking eligibility is set to at least 20 unique US orders for stability; this threshold is configurable in Streamlit.
- Lead-time, delay-frequency, and route-efficiency outputs are marked diagnostic until the source date issue is resolved.